



Innovative Paths for Digital Finance Empowerment: An Empirical Study of Integrating Intelligent and Green Manufacturing



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Abstract: Under the objective of establishing a manufacturing powerhouse, promoting deep integration of intelligent and green manufacturing has become a key initiative for achieving transformation and upgrading of the manufacturing industry. Exploring the internal logic of digital finance for executing such an integration is of paramount importance for the high-quality development of China's economy. While provincial panel data from 2011 to 2023 were collected as research samples, this paper employed a two-way fixed effects model to systematically and empirically examine the impact of digital finance on the proposed integration. The results demonstrated that digital finance formed a positive synergistic mechanism with environmental regulations and foreign direct investment to amplify its effect in propelling the integration of intelligent and green manufacturing. The transformation of scientific achievements and technological innovations also serves as the strategic propelling force in the integration process. This study provided empirical evidence and policy references for leveraging digital financial tools, improving the multi-policy synergy system, and accelerating the integration of intelligent and green manufacturing to achieve the “dual carbon” goals.

Keywords: Digital finance; Manufacturing industry; Integration of intelligence and greenization; Two-way fixed effects model

JEL Classification: G21, G32, O14, O33, Q50, Q55

1. Introduction

The transformation and upgrading of the manufacturing sector have ascended to a strategic imperative concerning national competitiveness and sustainable development. The report to the 20th National Congress of the Communist Party of China explicitly pointed out the need to “focus on the development of the real economy” and “accelerate the construction of a new development paradigm”. Against this backdrop, the deep integration of intelligence and greenization, serving as the two core dimensions of manufacturing modernization, is not only an inevitable choice to address global industrial competition but also a fundamental path to achieving “dual carbon” goals and resolving resource and environmental constraints.

The promulgation of “Made in China 2025” established the main direction of attack for the manufacturing industry, marking China's shift from factor-driven to innovation-driven development, and from scale expansion to quality and efficiency improvement. However, this transition process urgently requires efficient resource allocation mechanisms and the “living water” of finance. The rise of digital finance, characterized by high permeability, strong substitutability, and broad synergy, has broken the spatiotemporal limitations of traditional finance, providing new possibilities for resolving information asymmetry in the transformation and upgrading of manufacturing.

Under the guidance of the “Opinions on Accelerating the Construction of a Unified National Market”, guiding high-end factors such as capital and technology to cluster in advanced and green manufacturing has become a key link in deepening supply-side structural reform. Based on this, the current paper constructed a two-way fixed effects model from the theoretical perspective of total factor productivity enhancement and optimization of

industrial structure to empirically test the mechanism of digital finance in empowering the deep integration of intelligence and greenization in manufacturing. This study helped clarify the internal logic of the mutual promotion between the digital economy and the real economy, thus providing important decision-making references for promoting high-quality urban economic transformation and achieving comprehensive green and low-carbon development.

2. Literature Review

2.1 Literature Review on Digital Finance

As a new form of financial service in the digital era, digital finance profoundly influences the developmental path of the manufacturing industry through multiple mechanisms such as technological empowerment, capital allocation, and information integration. Gulati et al. (2026) pointed out through bibliometric analysis that research on digital finance had witnessed explosive growth since 2020. Especially in the wake of COVID-19, accelerated technological convergence has established digital finance as a key engine for industrial transformation.

From the perspective of IT governance, Almaqtari et al. (2025) commented that transformation of digital finance played a fully mediating role between IT governance and sustainable economic development. This indicates that digital finance is not merely a technical tool, but also a manifestation of institutional and governance capabilities. Xu et al. (2025) further highlighted that digital finance significantly enhanced the resilience of industrial chain, particularly by optimizing resource allocation through the mediating and moderating effects of technology enterprise incubation and regional innovation capability.

Meanwhile, Ma et al. (2025), approaching from the perspective of green technology innovation, discovered that digital finance indirectly promoted green innovation by alleviating financial mismatch and optimizing governance mechanisms.

In summary, the empowerment logic of digital finance is reflected not only in improved capital accessibility, but more importantly, in its systematic impact on governance structures, innovation ecosystems, and industrial chain resilience.

2.2 Literature Review on the Deep Integration of Smart and Green Manufacturing

The integration of intelligent and green manufacturing is a critical pathway to achieving high-quality development. Thurner & Roud (2016), based on empirical research involving over 600 eco-innovative enterprises in Russia, discovered that highly innovative and high-energy-consuming enterprises comprehensively improved green practices at both product and process levels, with effects extended to the entire supply chain. In contrast, state-owned enterprises focus primarily on enhancing resource efficiency and this reflects that green transformation has not yet formed a universal intrinsic linkage with intelligentization. Flachaire et al. (2014) and Kavre et al. (2026), in their study on small and medium manufacturing enterprises in emerging markets, indicated that although digitalization was regarded as key to sustainable growth, most enterprises lacked strategic roadmaps, capabilities of change management, and knowledge-sharing mechanisms to prevent the synergistic advancement of digitalization and greenization.

Muneer et al. (2026) further corroborated this from the perspective of green financing and found that while green financing could promote digital transformation and the adoption of circular economies in small and medium enterprises (SMEs), the prerequisite was that enterprises possessed strategic willingness and organizational readiness. In reality, most SMEs remain constrained by funding, technology, and governance structures. Jia (2026) introduced ergonomics into the discussion and proposed that the synergistic development of digitalization and greenization had to balance worker safety with energy efficiency. The scholar emphasized integrated optimization across multidimensional indicators.

In summary, the intelligent and green transformation of the manufacturing industry is still in its initial stage of synergy, facing multiple challenges such as technological, institutional, and regional heterogeneity.

2.3 Literature Review on Digital Finance Enabling the Synergistic Integration of Smart and Green Manufacturing

The way digital finance specifically promotes the deep integration of intelligent and green manufacturing is a core topic of the current research. Chen et al. (2024) and Claessens (2006) revealed that digital finance could directly promote green technology innovation in manufacturing enterprises and indirectly functioned by alleviating financing constraints and increasing research and development (R & D) investment, with corporate Environmental Social Governance (ESG) performance exhibiting a significant threshold effect. Interestingly, Wang (2024) pointed out that digital finance, through technological means such as big data and artificial intelligence, drove innovation, industrial upgrading, green transformation, and economic efficiency improvement in manufacturing

firms.

Sheng et al. (2023) and Yin (2022) further revealed that digital finance promoted technological innovation in manufacturing enterprises by alleviating financing constraints, and that the degree of market competition exhibited a double threshold effect. Zhou & Wen (2021), approaching from the path of intelligent transformation, identified three pathways including “collaborative R & D and processing”, “external factor driving”, and “value chain climbing”, wherein intelligent technology innovation and investment were necessary conditions.

These studies indicated that through multiple pathways such as financing alleviation, R & D incentives, governance optimization, and technological innovation, digital finance has become a key link connecting intelligentization and greenization.

2.4 Commentary on the Literature

Existing research has made significant progress in understanding how digital finance empowers manufacturing transformation; however, three main limitations remain. First, most studies focused on a single dimension of transformation, lacking a systematic examination of the deep integration of intelligent and green manufacturing. Second, there is a lack of in-depth discussions on spatial heterogeneity and regional coordination mechanisms. Third, the construction of holistic indicator systems requires further improvement. To address the research gaps found in these three aspects, this paper utilized provincial panel data from 2011 to 2023 in China to construct a two-way fixed effects model for further breakthroughs. First, regarding the indicator system, we constructed a comprehensive evaluation system covering both intelligent and green dimensions. Second, in terms of research perspective, we focused on the deep integration process of intelligent and green manufacturing, overcoming the limitations of single-transformation perspectives. Finally, in mechanism analysis, we deeply explored the spatial heterogeneity and regional synergistic effects of digital finance empowerment, to provide theoretical support and policy implications for the high-quality development of the manufacturing industry.

3. Theoretical Analysis and Development of Hypothesis

3.1 Impact of Digital Finance on the Deep Integration of Intelligence and Greenization in Manufacturing

As a product derived from the incorporation of digital technology and traditional finance, digital finance is profoundly altering the resource allocation models and forms of production function in the manufacturing sector. Its significant role could empower the deep integration of intelligence and greenization in manufacturing.

From the micro-level perspective of internal enterprise dynamics, Gao (2024) and Peng et al. (2025) proposed that digital finance directly addressed the bottlenecks of manufacturing transformation and upgrading by constructing a trinity of “capital-technology-policy” synergistic driving systems. In the process of transitioning towards intelligence and greenization, traditional manufacturing industries often face high research and development (R & D) costs and challenging financing constraints. With an objective to tackle these difficulties, Zuo & Hu (2026) leveraged frontier technologies such as big data and blockchain in digital finance to dismantle the spatiotemporal limitations of traditional financial paradigms, hence substantially enhancing the breadth of coverage and depth of usage of financial services. This not only effectively alleviated corporate financing constraints to provide necessary innovative financial support for intelligent manufacturing and low-carbon transformation, but also generated strong endogenous momentum through transmission mechanisms that improved production efficiency and increased government innovation subsidies. Meanwhile, Cao et al. (2026) emphasized that enterprises could utilize these funds to elevate investment in technological and human resources, thereby achieving a leap in production efficiency and an improvement in green total factor productivity at the micro level.

From the meso-level perspective of resource allocation, Cao et al. (2026) and Wang (2024) concluded that digital finance provided critical intermediary support for the high-quality development of manufacturing by reshaping the market-based allocation mechanism of production factors. Digital transformation is not merely a technological iteration but a profound revolution in resource allocation. To support this argument, Wang & Liao (2024) reported that the development of digital finance could significantly improve and reinforce the dominant role of the market in resource allocation. By raising the level of marketization, this progress guided the agglomeration of factors such as capital, technology, and talent towards advanced manufacturing sectors characterized by high efficiency and rich technological content. The improved efficiency in resource allocation served as a vital intermediary mechanism driving the transition of manufacturing from traditional extensive growth to high-quality development. It enabled factors to emanate from inefficient sectors and be precisely matched to projects involving intelligent manufacturing and green innovation, thereby promoting high-quality development in advanced manufacturing enterprises at the industrial level. Based on the aforementioned research, we proposed the following hypothesis.

H1: Digital finance has a significantly positive driving effect on the integration of intelligence and greenization in the manufacturing industry.

3.2 Transmission Mechanisms of Digital Finance Empowering the Deep Integration of Intelligence and Greenization in Manufacturing

The empowerment of digital finance on the integration of intelligence and greenization in manufacturing is, in essence, a dual revolution in industrial technological and ecological paradigms, triggered by the digital allocation of financial resources. This process entails more than a unidimensional capital injection; rather, it reshapes the dynamic mechanism of manufacturing transformation and upgrading by constructing a composite transmission system of “digital finance—technological innovation—environmental regulation and foreign direct investment (FDI) synergy”.

First, Chen (2023) and Zhou et al. (2026) pointed out that technological innovation was the core driver of industrial development and the key to the high-quality transformation of manufacturing, with funding for science and technology serving as a robust guarantee for such innovation. Chen (2023) observed that by leveraging tools such as big data and artificial intelligence, digital finance fully excavated potentially valuable information regarding the development of innovative enterprises. This effectively alleviated the high R & D costs and severe financing constraints that manufacturing enterprises faced under traditional financial paradigms. Such financial support was directly translated into corporate R & D momentum, to ensure a complete innovation chain from basic research to applied development, leading to production efficiency and improvement of green total factor productivity.

Secondly, the development of digital finance has constituted significant interactive empowerment effects with external environmental factors. On the one hand, Fan & Qu (2026) indicated that environmental regulation forced enterprises to bear higher pollution emission costs, compelling them to utilize digital finance resources for green technology upgrading to meet compliance requirements. Consequently, this achieved substantive improvements across multiple dimensions, including enhanced resource utilization efficiency, reduced pollutant emission intensity, and improved product greenization levels. On the other hand, the technology spillover effect brought by FDI was amplified under the catalysis of digital finance. Lyu (2023) and Shen et al. (2025) discovered that attracting foreign investment not only helped improve domestic digital economic strength and technological levels but also played a key driving role in industrial structure adjustment, upgrading, and the development of high-tech industries. Wu (2026) discovered that through optimizing resource allocation and guiding capital to precisely flow into advanced manufacturing sectors with high technological content, digital finance enabled the deep integration of the “forcing mechanism” of environmental regulation and the “technology spillover effect” of FDI, jointly promoting the high-quality transformation of manufacturing towards the deep integration of intelligence and greenization.

Finally, Jacobides et al. (2018) initially found that regional economic foundations and industrial structure gradients significantly moderate the empowerment effect, while Jacobides et al. (2024) further emphasized the critical role of digital infrastructure perfection, together revealing complex spatiotemporal heterogeneity. Weerasinghe et al. (2025) confirmed these patterns. Moreover, Masucci et al. (2020) and Xiao et al. (2026) pointed out that gaps in technological innovation, openness to the outside world, and industrial structure upgrading were the decisive forces affecting the development gap in manufacturing intelligence.

The empowerment effect of digital finance varies significantly across different regions and time periods. This variation depends not only on the accessibility of financial resources but, more importantly, on the perfection of the regional innovation system and the compatibility with the industrial structure. Only when digital finance forms a precise match with capabilities of regional technological innovation, environmental policy orientation, and the level of openness to the outside world could its driving effect be maximized on the integration of intelligence and greenization in manufacturing. Based on the aforementioned research, this paper also proposed the following two hypotheses.

H2: Digital finance primarily empowers the deep integration of intelligence and greenization in manufacturing through technological innovation.

H3: Digital finance forms a synergistic transmission mechanism with environmental regulation and foreign capital introduction, to promote the deep integration of intelligence and greenization in manufacturing.

4. Research Design

4.1 Model Construction and Variable Selection

4.1.1 Model specifications

This study employed a two-way fixed effects model to construct the empirical framework, controlling both individual and time effects. The specific calculation formula is as follows:

$$IGII_{it} = \alpha + \beta \times DIFI_{it} + \gamma \times X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

In the model, $IGII_{it}$ denotes the intelligent and green integration index of the manufacturing industry; $DIFI_{it}$ represents the digital inclusive finance index; X_{it} refers to control variables; μ_i indicates the individual fixed

effect; λ_t stands for the year fixed effect; and ε_{it} is the random disturbance term.

4.1.2 Variable definitions

(a) Dependent variable: Integration index of intelligence and greenization in manufacturing

The evaluation of intelligence centred on three core elements: talent, technology, and data. Meanwhile, the greenization evaluation indicators were selected to cover three key areas: pollution emission, energy utilization, and environmental protection (Table 1). To thoroughly assess the integration of intelligence and greenization in manufacturing, this study applied the Entropy Weight Method to conduct a composite measurement across these two dimensions, thereby calculating the Intelligence and Greenization Integration Index (IGII). The specific procedures included data standardization, calculation of indicator proportions, entropy value calculation, determination of difference coefficients, weight assignment, and the computation of the composite index.

Table 1. Evaluation index system for intelligent and green integration of the manufacturing industry

First-Level Index	Second-Level Index	Third-Level Index	Source
Intelligentization	Basic input	Talent input (person)	Meng et al. (2024); Xie et al. (2025)
		Internet popularization rate (%)	
		Installation density of industrial robots (unit)	
	Production application	Software Business Income (10,000 yuan)	Liu et al. (2024)
		Number of mobile phone subscribers at year-end (household)	
	Market benefit	Ratio of internet-related employees to employed population (%)	Yu (2025)
		Proportion of enterprises with e-commerce transactions (%)	
E-commerce transaction volume (100 million yuan/person)			
Greenization	Pollution emission	Mobile phone popularization rate (%)	Li & Qiu (2025)
		Industrial CO ₂ emissions (ton)	
		Industrial wastewater discharge (ton)	
	Energy utilization	General industrial solid waste generation (ton)	Qiu et al. (2024)
		Sulfur dioxide emissions	
		Nitrogen oxides emissions	
	Environmental protection	Chemical oxygen demand (COD) emissions	Tian & Ding (2019)
Investment in environmental pollution treatment (10,000 yuan)			
Industrial wastewater treatment intensity (10,000 yuan/ton)			
		Hazard-free treatment rate of domestic waste (%)	

(b) Explanatory variable: Digital financial inclusion index

This study constructed a three-tier indicator system for digital financial inclusion based on three dimensions: the breadth of coverage (lnbreadth), the depth of usage (lndepth), and the degree of digitalization (ln dig). Based on this system, the Digital Financial Inclusion Index was calculated and used as a proxy for the development level of digital financial inclusion in each region.

(c) Control variables: Environmental regulation intensity (ERI), foreign investment level (FIL), domestic invention patent authorizations (DIPA), and innovation product expenditure (IPE). Specifically, ERI represents the optimal allocation of green resources, FIL denotes the level of foreign capital introduction, DIPA serves as an indicator of technological development, and IPE reflects the support for technological innovation. To ensure data stationarity, the natural logarithms of certain variables were taken.

4.2 Sources of the Data

This paper utilized provincial panel data from 31 provinces in China. The core explanatory variable, the digital finance development index, was sourced from the Peking University Digital Financial Inclusion Index (2011–2023). Data regarding the explained variables, namely the level of intelligent and green development in the manufacturing industry as well as other relevant variables, were obtained from the China Statistical Yearbook, China Science and Technology Statistical Yearbook, China City Statistical Yearbook, and China Labor Statistical Yearbook. Additional data sources included statistical yearbooks of various provinces and municipalities over the years, the National Bureau of Statistics, the Development Research Center of the State Council Information Network, and the China National Intellectual Property Administration.

4.3 Model Testing

When estimating the panel data model using Ordinary Least Squares (OLS), it is necessary to first determine whether to adopt the Fixed Effects (FE) model or the Random Effects (RE) model. To this end, this study carried out the Hausman test using Stata 18 software. The test results showed that the Hausman test statistic is χ^2 ($p <$

0.01), indicating that the null hypothesis of the random effects model was rejected. Therefore, the fixed effects model was selected. To examine the existence of time effects, year dummy variables were included in the Fixed Effects model, and a joint significance test was performed on all year dummies. The test result yielded a p -value of less than 0.01, which rejected the null hypothesis of “no time effect”, implying that the model indeed exhibited time effects. Consequently, a two-way fixed effects model controlling both time and individual effects was finally adopted.

4.4 Empirical Results and Analysis

4.4.1 Baseline regression

Table 2 presents the estimation results based on the two-way fixed effects model. Column (5) in Table 2 illustrates the regression design where explanatory and control variables were progressively incorporated to analyze the impact of Digital Financial Inclusion (DIFI) and Environmental Regulation (lnERI) on the Manufacturing Intelligence-Greening Integration Index (IGII).

In Column (1), the core explanatory variable DIFI was included. The coefficient was 2.953 and significant at the 1% level, confirming the directly positive empowerment of digital financial inclusion on the integration of manufacturing intelligence and greening, which validates Hypothesis H1.

In Columns (2) and (3), lnERI and lnFIL were added step by step. Although the coefficient of DIFI slightly decreased to 2.927 and 2.816, it remained significant at the 1% level. Meanwhile, lnERI and lnFIL were both significantly positive at the 5% level. This indicated that the “forcing effect” of environmental regulation and the “technology spillover effect” of foreign investment formed a synergistic drive with digital finance, jointly promoting the integrated development of manufacturing intelligence and greening. This validates Hypothesis H2.

Columns (4) and (5) in Table 2 further incorporated Domestic Invention Patent Authorizations (DIPA) and Innovation Product Expenditure (IPE). The coefficient of DIPA remained stable at 0.003–0.004 and was significant at the 1% level, emerging as a core driver in the regression model. This confirms that the transformation of scientific and technological achievements serves as a universal support for the integration of manufacturing intelligence and greening, thereby validating Hypothesis H3. In contrast, the coefficient of IPE was 0.016 but failed to pass the significance test. This insignificance may be attributed to two main factors: first, a structural misalignment between the allocation of innovation funds and the in-depth integration needs of the industry, as most funds are biased towards R & D in either intelligence or greening separately rather than their synergy, rendering it difficult to directly boost the integration index; and second, a time lag effect, where the empowerment of research funding requires a cycle of R & D, pilot testing, and implementation, meaning the current period effect has not yet manifested.

Table 2. Results of baseline regression

Variables	IGII	IGII	IGII	IGII	IGII
<i>DIFI</i>	2.953*** (5.591)	2.927*** (5.379)	2.816*** (5.524)	0.607 (1.494)	0.582 (1.340)
<i>lnERI</i>	—	11.273** (2.329)	10.769** (2.245)	8.168** (2.646)	7.379** (2.135)
<i>lnFIL</i>	—	—	8.783** (2.051)	2.391 (1.255)	3.133* (1.785)
<i>DIPA</i>	—	—	—	0.004*** (8.482)	0.003*** (7.589)
<i>IPE</i>	—	—	—	—	0.016 (0.767)
Constant	-611.947*** (-4.554)	-638.662*** (-4.538)	-633.464*** (-4.610)	-89.013 (-0.801)	88.158 (-0.760)
Observations	403	403	403	403	403
<i>R</i> -squared	0.881	0.886	0.888	0.972	0.973
Individual fixed effects	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES

Note: Robust t -statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; A em dash (—) denotes no data;
IGII = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

In 2015, the State Council issued the “Made in China 2025” strategy, which explicitly identified “smart manufacturing” as the main direction of attack while synchronously promoting green manufacturing to build an efficient, clean, low-carbon, and circular green manufacturing system. This initiative was accompanied by supporting policies on digital finance and environmental regulation standards. By strengthening the rigidity of

environmental regulations and guiding digital finance and foreign investment to tilt towards the integration sector, this policy aligns perfectly with the findings in Table 1, particularly about the synergistic drive of green resource allocation, foreign investment, and technological development. It directly promotes the “forcing effect” of environmental regulation and provides a policy orientation for digital finance empowerment and foreign investment technology spillovers, thereby corroborating the logical validity of the baseline regression model.

4.4.2 Robustness check

(1) Robustness check by excluding municipalities

Table 3 tests the robustness of the baseline regression conclusions by excluding the municipality samples, and the results indicated that the baseline regression possessed strong robustness. After the adjustment, the core explanatory variable DIFI and the control variables *lnERI*, *lnFIL*, and *DIPA* all remained significant, with signs consistent with the baseline regression. Notably, the coefficient of *IPE* increased to 0.06 and became significant at the 1% level, further corroborating the potential empowering value of research funding support. The core conclusions were not disturbed by the special sample of municipalities. This demonstrated that after excluding these municipalities, the impact of core driving factors on the integration of manufacturing intelligence and greening had not undergone fundamental changes, indicating that the baseline regression conclusions had reliable generalizability.

Table 3. Robustness test: Excluding municipalities

Variables	IGII	IGII
<i>DIFI</i>	0.582*** (2.757)	-0.252* (-1.688)
<i>lnERI</i>	7.379*** (3.520)	4.912*** (2.980)
<i>lnFIL</i>	3.133** (2.501)	1.953** (2.025)
<i>DIPA</i>	0.003*** (13.928)	0.002*** (4.125)
<i>IPE</i>	0.016* (1.720)	0.060*** (4.899)
Constant	-88.158 (-1.618)	112.911*** (3.234)
N	403	351
R-squared	0.973	0.984

Note: *t*-statistics in parentheses; **p* < 0.1, ***p* < 0.05, ****p* < 0.01; *IGII* = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

(2) Robustness check by excluding samples from the first and last years

Table 4. Robustness test: Excluding the first and last year observations

Variables	IGII	IGII
<i>DIFI</i>	0.582*** (2.757)	0.469** (2.269)
<i>lnERI</i>	7.379*** (3.520)	6.827*** (2.965)
<i>lnFIL</i>	3.133** (2.501)	2.257** (2.010)
<i>DIPA</i>	0.003*** (13.928)	0.003*** (8.808)
<i>IPE</i>	0.016* (1.720)	0.018 (1.471)
Constant	-88.158 (-1.618)	-59.169 (-1.079)
N	403	341
R-squared	0.973	0.975

Note: *t*-statistics in parentheses; **p* < 0.1, ***p* < 0.05, ****p* < 0.01; *IGII* = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

Table 4 excludes data from the first and last years of the sample to rule out the interference of initial data fluctuations and end-period policy shocks, yielding ideal robustness check results. Among the core variables, *DIFI*,

lnERI, lnFIL, and DIPA were all significantly positive, with signs completely consistent with the baseline regression. Although the coefficient of IPE slightly increased to 0.018 and failed to pass the significance test, its sign remains consistent with the baseline regression without any reverse change, indicating that the core driving effect has not shifted. This suggests that the baseline regression results are not driven by special data in the first and last years. By ruling out the interference of data outliers, it is confirmed that the core conclusions of the baseline regression model possess strong robustness.

4.4.3 Endogeneity test

Table 5 tests the potential endogeneity issues such as bidirectional causality and omitted variables. The results indicated that endogeneity had a limited impact on the core conclusions. After the endogeneity test, although the coefficient of the core explanatory variable DIFI decreased compared with the baseline regression, it stayed significant at the 5% level in one column, with the sign remaining consistently positive. Among the control variables, lnFIL, DIPA, and IPE all maintained positive significance, while the sign of lnERI stayed intact with only a slight adjustment in significance. This demonstrated that even when considering potential endogeneity, the positive driving relationship of core factors such as digital inclusive finance and technological innovation outcomes on the integration of manufacturing intelligence and greening still holds. This further validates hypothesis H1, indicating that the estimation results of the baseline regression have high credibility.

Table 5. Endogeneity test

Variables	IGII	IGII	IGII
<i>DIFI</i>	0.133** (2.308)	0.126 (1.574)	0.134 (1.636)
<i>lnERI</i>	6.441* (1.934)	5.993 (1.430)	6.286 (1.482)
<i>lnFIL</i>	5.619** (2.282)	5.827** (2.175)	5.939** (2.195)
<i>DIPA</i>	0.004*** (8.381)	0.004*** (8.370)	0.004*** (8.335)
<i>IPE</i>	0.030*** (2.750)	0.030*** (2.723)	0.030*** (2.712)
Constant	5.698 (0.203)	8.333 (0.219)	5.115 (0.133)
N	372	341	341
R-squared	0.931	0.930	0.930

Note: *t*-statistics in parentheses; **p* < 0.1, ***p* < 0.05, ****p* < 0.01; IGII = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

4.4.4 Heterogeneity analysis

(1) Regional heterogeneity analysis

Table 6 divides the national sample into four major regions, i.e., Eastern, Central, Western, and Northeastern to compare the regional differences in the impact of digital finance and control variables on the integration of manufacturing intelligence and greening. The regional heterogeneity of the core explanatory variable DIFI is significant: (i) the Central and Northeastern regions were significant at the 1% level, identifying them as the core beneficiaries of digital finance empowerment; (ii) the Eastern region showed marginal significance at the 10% level, indicating a marginalized empowerment effect; (iii) the coefficient for the Western region failed to pass the significance test, suggesting no obvious empowerment effect. The causes of this disparity are as follows: the Central region was currently undertaking industrial transfer, where the demand for digital finance precisely matched the needs of manufacturing upgrading; the Northeast region, as an old industrial base, relied on digital finance as a crucial support for its industrial transformation and upgrading; the integration of manufacturing intelligence and greening in the Eastern region had already reached a high level, so the empowerment effect of digital finance was less pronounced; and in the Western region, imperfect digital infrastructure and insufficient penetration prevented the full realization of empowerment potential of digital finance.

The regional differences in control variables are equally critical: lnERI was significantly positive in the Eastern, Central, and Western regions but insignificant in the Northeastern region, indicating that the “forcing effect” of environmental regulation had not yet manifested in the Northeast. lnFIL was significant only in the Eastern and Northeastern regions, suggesting that the Central and Western regions suffered from inappropriate quality of foreign capital introduction and limited technology spillover effects. DIPA was significantly positive across all regions; as the only core driving factor without regional variation, it validated the generalizability of innovation outcome transformation. IPE was significant only in the Western region and marginally significant in the Eastern region, but insignificant in the Central and Northeastern regions, reflecting the more urgent demand for innovation

funding in the West and the need for improved funding efficiency in the East. The DIFI coefficient in the full-sample regression was 0.582, representing the average regional effect. In contrast, the DIFI coefficient for the Eastern region was 0.893. Although marginally significant, this coefficient was higher than the average, suggesting that digital finance in the Eastern region still has room for empowerment. Yet, the bottleneck of diminishing marginal returns has to be overcome in the first priority.

Table 6. Regional heterogeneity analysis

Variables	IGII	IGII	IGII	IGII	IGII
<i>DIFI</i>	0.582*** (2.757)	0.893* (1.754)	0.750*** (2.766)	0.026 (0.175)	1.206*** (3.189)
<i>lnERI</i>	7.379*** (3.520)	12.567*** (2.729)	7.892*** (4.532)	4.500*** (2.692)	0.185 (0.106)
<i>lnFIL</i>	3.133** (2.501)	15.505** (2.520)	-0.125 (-0.045)	-0.417 (-0.363)	2.116** (2.192)
<i>DIPA</i>	0.003*** (13.928)	0.003*** (9.234)	0.002*** (2.731)	0.002*** (4.520)	0.004** (2.512)
<i>IPE</i>	0.016* (1.720)	0.020* (1.966)	0.008 (0.266)	0.078*** (3.919)	0.001 (0.033)
Constant	-88.158 (-1.618)	-215.962 (-1.499)	-114.490* (-1.920)	44.647 (1.280)	-223.133** (-2.525)
N	403	130	78	156	39
R-squared	0.973	0.965	0.975	0.957	0.987

Note: *t*-statistics in parentheses; **p* < 0.1, ***p* < 0.05, ****p* < 0.01; IGII = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

(2) Period heterogeneity analysis

Year 2016 served as a critical dividing point between the development stage of digital finance and the process of deep integration of manufacturing intelligence and greening. Prior to this, digital finance expanded rapidly and directly empowered the initial integration of manufacturing intelligence and greening. Subsequently, its marginal effect diminished, and the integration faced multiple bottlenecks. Therefore, taking 2016 as the time node, the sample was divided into two periods, 2011–2016 and 2017–2023, to examine the temporal differences in the empowerment effect of digital finance.

The temporal heterogeneity of the core explanatory variable *DIFI* is evident: the coefficient was significant at the 1% level during 2011–2016, while it was marginally significant at the 10% level during 2017–2023. Although both were significantly positive, the coefficient in the earlier period was larger and exhibited higher significance. The core reason for this disparity is as follows: prior to 2016, digital finance was in a stage of rapid development with obvious policy dividends, and the integration of manufacturing intelligence and greening was still in its initial stage, allowing digital finance to have a direct empowerment effect. After 2017, the penetration of digital finance increased, and the expanded base led to diminishing marginal effects. Meanwhile, the integration of manufacturing intelligence and greening faced multiple bottlenecks in technology and mechanisms, hence weakening the empowerment effect of relying solely on digital finance.

The temporal differences in control variables further revealed the evolution of the mechanism for the integration of manufacturing intelligence and greening: the coefficient of *lnERI* was significant at the 1% level in the earlier period and marginally significant at the 10% level in the later period, indicating that the “forcing effect” of environmental regulation was more direct initially, while the marginal impact of regulation weakened as the integration level improved. *lnFIL* was insignificant in the earlier period but significant at the 5% level in the later period, reflecting improved quality of foreign capital introduction and the gradual emergence of technology spillover effects in the later stage. *DIPA* remained significant at the 1% level throughout, serving as a core driving factor across the entire period; with a slightly higher coefficient in the later period, it suggests the continued prominence of the importance of innovation outcomes. *IPE* is significant at the 1% level in the earlier period but insignificant in the later period, possibly because the large funding gap in the earlier stage led to significant marginal effects, while funding saturation in the later stage was not accompanied by a synchronous improvement in usage efficiency.

The full-sample *DIFI* coefficient of 0.582 represents the average effect across periods, while the model’s *R*² was higher in the later period than in the earlier one, indicating better goodness-of-fit for the later sample and a more mature mechanism for digital finance to enhance the integration of manufacturing intelligence and greening. Currently, it is necessary to strengthen the synergy between digital finance and environmental regulation, foreign capital quality, and innovation efficiency. By optimizing resource allocation and improving the efficiency of innovation transformation, we could overcome the dilemma of diminishing marginal returns in digital finance

empowerment and promote the integration of manufacturing intelligence and greening toward a higher level of development. (Table 7)

Table 7. Period heterogeneity analysis

Variables	IGH	IGH	IGH
<i>DIFI</i>	0.582*** (2.757)	0.669*** (2.635)	0.494* (1.785)
<i>lnERI</i>	7.379*** (3.520)	11.862*** (4.484)	3.373* (1.672)
<i>lnFIL</i>	3.133** (2.501)	2.694 (1.436)	2.876** (2.141)
<i>DIPA</i>	0.003*** (13.928)	0.002*** (5.335)	0.003*** (11.910)
<i>IPE</i>	0.016* (1.720)	0.100*** (8.889)	-0.001 (-0.068)
Constant	-88.158 (-1.618)	-106.298*** (-2.611)	-65.714 (-0.701)
N	403	186	217
<i>R-squared</i>	0.973	0.980	0.987

Note: *t*-statistics in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; IGH = Intelligence and Greenization Integration Index; *DIFI* = Digital Financial Inclusion Index; *lnERI* = natural logarithm of Environmental Regulation Intensity; *lnFIL* = natural logarithm of Foreign Investment Level; *DIPA* = Domestic Invention Patent Authorizations; and *IPE* = Innovation Product Expenditure.

5. Conclusions and Recommendations

5.1 Conclusions

(1) Digital finance has a significantly positive driving effect on the integration of manufacturing intelligence and greening; it forms a synergistic empowerment effect with environmental regulation and foreign capital introduction. The coefficient of *DIFI* was significantly positive, directly promoting the improvement of the integration level. The green resource allocation effect of environmental regulation and the technology spillover effect of foreign capital introduction were both significantly positive, jointly promoting the integration of manufacturing intelligence and greening with digital finance.

(2) Technological innovation serves as a universal core driving force for the integration of intelligence and greening. In all regression models, the number of *DIPA* was consistently significantly positive with high coefficient stability. This validates that the transformation of scientific and technological achievements is the most universal driver for promoting the integration of manufacturing intelligence and greening. However, *IPE* failed to pass the significance test in the baseline regression and was only significant in some robustness checks, suggesting that current R & D investment may suffer from structural or temporal mismatch and fail to directly translate into the momentum for the integration of manufacturing intelligence and greening.

(3) The empowerment effect of digital finance exhibited significant regional heterogeneity. Sub-sample tests by region showed that the central and northeastern regions were the core empowerment areas for digital finance. While the integration of manufacturing intelligence and greening in the eastern region had reached a relatively high level, the empowerment effect of digital finance was not obvious, and the effect in the western region was insignificant. The effects of environmental regulation, foreign capital introduction, and innovation funding appeared with regional differentiation, whereas the transformation of scientific and technological achievements was the only core supporting factor with no regional differences.

(4) The empowerment effect of digital finance acknowledged significant temporal heterogeneity, and the driving mechanism was gradually shifting towards diversified synergy. Taking 2016 as the dividing point, the intensity and significance of digital finance empowerment were higher during 2011–2016, while the marginal effect diminished during 2017–2023. The “forcing effect” of environmental regulation was stronger in the earlier period, whereas the roles of foreign capital technology spillover and the transformation of innovation achievements have been continuously strengthened in the later period.

5.2 Recommendations

(1) Establish a digital finance service framework oriented towards the convergence of smart and green manufacturing. On the one hand, the government should steer financial resources including credit, equity, insurance, and supply chain finance to specifically target technological upgrades, modernization of green smart equipment, and low-carbon smart manufacturing projects that foster this convergence. Given the high density of small and medium-sized manufacturers, digital technologies should be leveraged to lower financing thresholds

and expand the coverage of credit and technical renovation loans, thereby enhancing the accessibility of digital finance. On the other hand, it is crucial to establish a performance evaluation mechanism for digital finance empowerment. By focusing on the efficiency of convergence output, this mechanism aims to prevent capital misallocation and inefficient deployment, thus effectively revitalizing existing financial resources.

(2) Implement differentiated and flexible environmental regulation policies. On the one hand, while reinforcing rigid constraints, the government should deploy supporting measures such as green technological renovation subsidies and carbon emission reduction support tools. This “pressure-and-incentive” dual mechanism is designed to compel and encourage manufacturing enterprises to proactively pursue convergent transformation. On the other hand, it is essential to improve the quality of foreign investment introduction. Priority should be given to technology-intensive foreign projects in green and smart manufacturing to strengthen technology spillovers and local absorption capabilities, thereby aligning the manufacturing sector with advanced international convergence standards. Ultimately, this paves the way for establishing a synergistic platform that integrates digital finance, environmental regulations, and foreign investment policies.

(3) Prioritize invention patents related to the convergence of smart and green manufacturing as a focal point for supporting technological innovation. On the one hand, the government should optimize the allocation structure of innovation funding. This involves reducing investment in single-technology R & D and prioritizing cross-disciplinary and convergent innovation projects. Furthermore, establishing a fund tracking and performance evaluation system is crucial to mitigate the issue of lagging investment returns. On the other hand, leveraging digital economy and industrial innovation platforms, the government should facilitate the joint establishment of convergent technology R & D centers by universities, research institutes, and manufacturing enterprises, thereby enhancing independent supply capabilities.

(4) Establish and promote demonstration zones for the integration of digital finance and manufacturing. On the one hand, explore replicable and scalable models to extend digital financial services to county-level industrial clusters and niche sectors of traditional manufacturing. This initiative aims to bridge the gap in digital transformation and green upgrading for SMEs. On the other hand, strengthen inter-provincial coordination to align with the needs of industrial upgrading in central and northeastern regions, thereby expanding the scope of digital finance to extended financing across the chains of the manufacturing industry.

Author Contributions

Conceptualization, X.H.D.; methodology, X.H.D.; writing—review and editing, X.H.D.; data curation, X.Y.S.; formal analysis, X.Y.S.; writing—original draft preparation, X.Y.S. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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